

Laplace Transforms with Non-Zero Initial Conditions and Boundary Values

A Self-Study Reference & Master Guide

Contents

Part I — Theory & Worked Examples	1
2.10 The $t = 0$ Trap — Why the Standard Rule Needs Data at Zero	1
2.11 The “Clock Reset” Substitution for Shifted IVPs	2
2.12 Boundary Value Problems — Parameterizing the Missing Data	4
Part II — Practice Set	8
Part III — Complete Solutions	9

Part I — Theory & Worked Examples

Non-Zero Conditions at a Glance: Core Sub-Topics

- **The Hard-Wired Lower Bound (2.10):** the Laplace integral starts at $t = 0$, and nothing can move it. The derivative rule $\mathcal{L}\{y'\} = sY(s) - y(0)$ feeds on the value *at zero* — so a condition given anywhere else cannot enter the transform directly.
- **The “Clock Reset” Substitution (2.11):** for an IVP whose data live at $t = a \neq 0$, slide the time axis with $\tau = t - a$. The shifted problem has its conditions at $\tau = 0$, exactly where Laplace can use them. Solve in τ , then slide back.
- **Parameterizing Boundary Value Problems (2.12):** when one condition sits at $t = 0$ and another at $t = b$, the missing initial slope is unknown. Carry it as a symbol c , solve the ODE in terms of c , then pin c down with the far boundary condition.

2.10 The $t = 0$ Trap — Why the Standard Rule Needs Data at Zero

Every Laplace-based ODE solve rests on the derivative rule

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0), \quad \mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0).$$

These constants $y(0)$ and $y'(0)$ are not a stylistic choice — they fall directly out of integration by parts on the *defining* integral, whose lower limit is permanently fixed at zero:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt.$$

The boundary term from integration by parts is evaluated at that lower limit, which is why the rule asks for the value *at* $t = 0$ and nowhere else.

The $t = 0$ Trap

The derivative rule $\mathcal{L}\{y'\} = sY(s) - y(0)$ is **hard-wired to consume the value at zero**. The lower bound of the Laplace integral is fixed at 0; it is part of the definition and cannot be slid to another point.

So if you are handed a condition such as $y(2) = 5$ — a value at $t = 2$ — there is **no slot** for it in the transform. Writing “ $sY(s) - y(2)$ ” is meaningless: the rule does not have a $y(2)$ term, and inventing one corrupts the algebra. The transform only ever knows $y(0)$ and $y'(0)$.

The fix depends on what you were given:

- All conditions sit at one common point $t = a \neq 0$ (a shifted IVP) $\Rightarrow$ slide the clock so that point becomes $\tau = 0$ (Section 2.11).
- Conditions are *split* between $t = 0$ and some $t = b$ (a boundary value problem) $\Rightarrow$ parameterize the unknown zero-data as a constant c (Section 2.12).

2.11 The “Clock Reset” Substitution for Shifted IVPs

When all of the initial data are stacked at the same point $t = a$ (for example $y(a)$ and $y'(a)$), the cure is to relabel time so that a becomes the new origin. This is a pure change of variable — it does not alter the differential equation, only the name of the independent variable — and it places the data exactly where the transform expects them.

The “Clock Reset” Substitution — Four Steps

Given an IVP whose conditions are specified at $t = a$:

Step 1 — Reset the clock. Define the shifted time

$$\tau = t - a, \quad \text{equivalently} \quad t = \tau + a.$$

When $t = a$ the new clock reads $\tau = 0$, so the data move to the origin.

Step 2 — Translate the derivatives (chain rule). Write $y(t) = y(\tau + a)$ as a function of τ . Since $\frac{d\tau}{dt} = \frac{d}{dt}(t - a) = 1$, the chain rule gives

$$\frac{dy}{dt} = \frac{dy}{d\tau} \cdot \frac{d\tau}{dt} = \frac{dy}{d\tau} \cdot 1 = \frac{dy}{d\tau}.$$

Differentiating once more, again with $\frac{d\tau}{dt} = 1$,

$$\frac{d^2y}{dt^2} = \frac{d}{dt} \left(\frac{dy}{d\tau} \right) = \frac{d}{d\tau} \left(\frac{dy}{d\tau} \right) \frac{d\tau}{dt} = \frac{d^2y}{d\tau^2}.$$

Because the shift factor is exactly 1, *every derivative is unchanged in form*: $y'(t) = y'(\tau)$ and $y''(t) = y''(\tau)$. The ODE keeps its coefficients verbatim.

Step 3 — Solve entirely in the τ domain. The problem now reads with conditions at $\tau = 0$, where $\mathcal{L}\{y'\} = sY - y(0)$ applies cleanly. Transform, solve for $Y(s)$, and invert to get $y(\tau)$.

Step 4 — Slide the clock back. Undo the substitution with $\tau = t - a$, replacing every τ in the final answer to recover $y(t)$.

Worked Example 1: A Shifted Second-Order IVP

Solve $y'' + y = 0$ with $y(\frac{\pi}{2}) = 1$ and $y'(\frac{\pi}{2}) = 0$.

Step 1 — Reset the clock. The data sit at $a = \frac{\pi}{2}$, so let

$$\tau = t - \frac{\pi}{2}, \quad t = \tau + \frac{\pi}{2}.$$

At $t = \frac{\pi}{2}$ we have $\tau = 0$, so the conditions become $y(0) = 1$ and $y'(0) = 0$ in the τ clock.

Step 2 — Translate the ODE. By the chain rule with $\frac{d\tau}{dt} = 1$, derivatives carry over unchanged, so the equation is identical in form:

$$y''(\tau) + y(\tau) = 0, \quad y(0) = 1, \quad y'(0) = 0.$$

Step 3 — Transform. Apply $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$ with $y(0) = 1$, $y'(0) = 0$:

$$s^2Y(s) - s(1) - 0 + Y(s) = 0 \implies (s^2 + 1)Y(s) = s.$$

Step 4 — Solve for $Y(s)$.

$$Y(s) = \frac{s}{s^2 + 1}.$$

Step 5 — Invert. Matching $\frac{s}{s^2 + 1} \rightarrow \cos \tau$:

$$y(\tau) = \cos \tau.$$

Step 6 — Slide the clock back. Replace $\tau = t - \frac{\pi}{2}$:

$$y(t) = \cos\left(t - \frac{\pi}{2}\right) = \sin t.$$

The last step uses the identity $\cos(\theta - \frac{\pi}{2}) = \sin \theta$. A quick check confirms it: $\sin(\frac{\pi}{2}) = 1$ matches $y(\frac{\pi}{2}) = 1$, and $y'(t) = \cos t$ gives $\cos(\frac{\pi}{2}) = 0$, matching $y'(\frac{\pi}{2}) = 0$.

Key Takeaway

For a shifted IVP, the substitution $\tau = t - a$ is a rigid translation of the time axis. Because $\frac{d\tau}{dt} = 1$, the derivatives — and therefore the ODE — are untouched; only the *location* of the data moves, landing at $\tau = 0$ where Laplace can finally use it.

2.12 Boundary Value Problems — Parameterizing the Missing Data

A boundary value problem (BVP) hands you conditions at *two different* points — typically one at $t = 0$ and one at a far endpoint $t = b$. The clock-reset trick cannot rescue this case: sliding the axis to put one condition at the origin only pushes the other condition off the origin. We need a different idea.

The transform demands a complete set of data at $t = 0$: for a second-order equation it wants

both $y(0)$ and $y'(0)$. A BVP usually gives one of them and withholds the other. The resolution is to **name the missing piece**.

Parameterizing a Boundary Value Problem

Given a second-order BVP with one condition at $t = 0$ and one at $t = b$:

Step 1 — Name the unknown zero-data. Suppose $y(0)$ is given but $y'(0)$ is not. Introduce a symbolic constant

$$y'(0) = c,$$

treating it as a known value for now. (If instead $y(0)$ is the missing one, name *that* c .)

Step 2 — Transform with the symbol carried along. Apply $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$, leaving c in place. Solve for $Y(s)$ as an expression that contains c .

Step 3 — Invert to a general solution in c . The inverse transform produces $y(t)$ as a family of solutions parameterized by the single unknown c .

Step 4 — Apply the far boundary condition. Substitute the second condition (at $t = b$) into $y(t)$ — or into $y'(t)$, whichever the condition specifies. This yields one linear equation in the single unknown c . Solve it, and back-substitute to get the explicit solution.

Worked Example 2: A Mixed Boundary Value Problem

Solve $y'' - y = 0$ with $y(0) = 1$ and $y'(2) = 1$.

Step 1 — Name the missing data. We know $y(0) = 1$ but not $y'(0)$. Set

$$y'(0) = c.$$

Step 2 — Transform, carrying c . With $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0) = s^2Y - s - c$:

$$s^2Y(s) - s - c - Y(s) = 0 \implies (s^2 - 1)Y(s) = s + c.$$

Step 3 — Solve for $Y(s)$ and split.

$$Y(s) = \frac{s + c}{s^2 - 1} = \frac{s}{s^2 - 1} + \frac{c}{s^2 - 1}.$$

Step 4 — Invert. Using $\frac{s}{s^2 - 1} \rightarrow \cosh t$ and $\frac{1}{s^2 - 1} \rightarrow \sinh t$:

$$y(t) = \cosh t + c \sinh t.$$

This is the general solution, still carrying the unknown c .

Step 5 — Differentiate for the boundary condition. The condition is on y' , so differentiate, using $\frac{d}{dt} \cosh t = \sinh t$ and $\frac{d}{dt} \sinh t = \cosh t$:

$$y'(t) = \sinh t + c \cosh t.$$

Step 6 — Apply $y'(2) = 1$ and solve for c . Substitute $t = 2$:

$$y'(2) = \sinh 2 + c \cosh 2 = 1 \implies c \cosh 2 = 1 - \sinh 2.$$

Isolate c by dividing by $\cosh 2$:

$$c = \frac{1 - \sinh 2}{\cosh 2} \approx \frac{1 - 3.6269}{3.7622} \approx -0.6982.$$

Step 7 — Write the explicit solution. Back-substitute the value of c :

$$y(t) = \cosh t + \frac{1 - \sinh 2}{\cosh 2} \sinh t.$$

By construction $y(0) = \cosh 0 = 1$, and c was chosen precisely so that $y'(2) = 1$ — both boundary conditions are satisfied.

Key Takeaway

A BVP withholds part of the $t = 0$ data the transform requires. Name the missing value c , solve the ODE as a one-parameter family $y(t; c)$, then let the far boundary condition supply the single equation that fixes c . The Laplace machinery runs untouched; the only new work is one linear solve at the end.

Part II — Practice Set

Work each problem before consulting the complete solutions in Part III. Decide first which tool applies: a single shifted data point calls for the clock reset ($\tau = t - a$); split conditions call for parameterizing the missing zero-data with a constant.

Problem 1: First-Order Shifted IVP.

Solve

$$y' + 2y = 0, \quad y(3) = 4.$$

Hint: the data sit at $t = 3$. Reset the clock with $\tau = t - 3$.

Problem 2: Second-Order Shifted IVP.

Solve

$$y'' + 4y = 0, \quad y(\pi) = 2, \quad y'(\pi) = -4.$$

Hint: both conditions sit at $t = \pi$. Reset with $\tau = t - \pi$, then simplify using periodicity at the end.

Problem 3: Mixed Boundary Value Problem.

Solve

$$y'' + 9y = 0, \quad y(0) = 2, \quad y\left(\frac{\pi}{6}\right) = 1.$$

Hint: $y(0)$ is given but $y'(0)$ is not. Let $y'(0) = c$ and use the far condition to solve for it.

Part III — Complete Solutions

Solution 1: First-Order Shifted IVP.

The single data point lives at $t = 3$, so we reset the clock.

Step 1 — Reset the clock. Let

$$\tau = t - 3, \quad t = \tau + 3.$$

At $t = 3$ the new clock reads $\tau = 0$, so the condition becomes $y(0) = 4$.

Step 2 — Translate the ODE. Since $\frac{d\tau}{dt} = 1$, the chain rule gives $y'(t) = y'(\tau)$, so the equation is unchanged in form:

$$y'(\tau) + 2y(\tau) = 0, \quad y(0) = 4.$$

Step 3 — Transform. Apply $\mathcal{L}\{y'\} = sY - y(0)$ with $y(0) = 4$:

$$sY(s) - 4 + 2Y(s) = 0 \implies (s + 2)Y(s) = 4.$$

Step 4 — Solve for $Y(s)$.

$$Y(s) = \frac{4}{s + 2}.$$

Step 5 — Invert. Using $\frac{1}{s + 2} \rightarrow e^{-2\tau}$:

$$y(\tau) = 4e^{-2\tau}.$$

Step 6 — Slide the clock back. Replace $\tau = t - 3$:

$$y(t) = 4e^{-2(t-3)}.$$

Check: $y(3) = 4e^0 = 4$, as required.

Solution 2: Second-Order Shifted IVP.

Both conditions sit at $t = \pi$, so a single clock reset places them at the origin.

Step 1 — Reset the clock. Let

$$\tau = t - \pi, \quad t = \tau + \pi.$$

The conditions become $y(0) = 2$ and $y'(0) = -4$ in the τ clock.

Step 2 — Translate the ODE. With $\frac{d\tau}{dt} = 1$, both derivatives carry over unchanged ($y'(t) = y'(\tau)$, $y''(t) = y''(\tau)$), so

$$y''(\tau) + 4y(\tau) = 0, \quad y(0) = 2, \quad y'(0) = -4.$$

Step 3 — Transform. Apply $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$ with $y(0) = 2$, $y'(0) = -4$:

$$s^2Y(s) - 2s - (-4) + 4Y(s) = 0 \implies (s^2 + 4)Y(s) = 2s - 4.$$

Step 4 — Solve for $Y(s)$ and split.

$$Y(s) = \frac{2s-4}{s^2+4} = 2 \cdot \frac{s}{s^2+4} - 2 \cdot \frac{2}{s^2+4}.$$

The second term is written with a 2 in the numerator to match the standard form $\frac{2}{s^2+2^2} \rightarrow \sin(2\tau)$.

Step 5 — Invert. Using $\frac{s}{s^2+4} \rightarrow \cos(2\tau)$ and $\frac{2}{s^2+4} \rightarrow \sin(2\tau)$:

$$y(\tau) = 2\cos(2\tau) - 2\sin(2\tau).$$

Step 6 — Slide the clock back. Replace $\tau = t - \pi$:

$$y(t) = 2\cos(2(t-\pi)) - 2\sin(2(t-\pi)).$$

Now simplify using $2(t-\pi) = 2t - 2\pi$ and the 2π -periodicity of sine and cosine, so $\cos(2t - 2\pi) = \cos 2t$ and $\sin(2t - 2\pi) = \sin 2t$:

$$y(t) = 2\cos(2t) - 2\sin(2t).$$

Check: $y(\pi) = 2\cos(2\pi) - 2\sin(2\pi) = 2$, and $y'(t) = -4\sin(2t) - 4\cos(2t)$ gives $y'(\pi) = -4\cos(2\pi) = -4$. Both conditions hold.

Solution 3: Mixed Boundary Value Problem.

The conditions are split between $t = 0$ and $t = \frac{\pi}{6}$, so we parameterize the missing initial slope.

Step 1 — Name the missing data. We have $y(0) = 2$ but not $y'(0)$. Set

$$y'(0) = c.$$

Step 2 — Transform, carrying c . With $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0) = s^2Y - 2s - c$:

$$s^2Y(s) - 2s - c + 9Y(s) = 0 \implies (s^2 + 9)Y(s) = 2s + c.$$

Step 3 — Solve for $Y(s)$ and split.

$$Y(s) = \frac{2s+c}{s^2+9} = 2 \cdot \frac{s}{s^2+9} + \frac{c}{3} \cdot \frac{3}{s^2+9}.$$

The factor of $\frac{1}{3}$ appears so the last term matches $\frac{3}{s^2+3^2} \rightarrow \sin(3t)$.

Step 4 — Invert. Using $\frac{s}{s^2+9} \rightarrow \cos(3t)$ and $\frac{3}{s^2+9} \rightarrow \sin(3t)$:

$$y(t) = 2\cos(3t) + \frac{c}{3}\sin(3t).$$

Step 5 — Apply the far boundary condition. Substitute $t = \frac{\pi}{6}$, noting $3 \cdot \frac{\pi}{6} = \frac{\pi}{2}$, so $\cos(\frac{\pi}{2}) = 0$ and $\sin(\frac{\pi}{2}) = 1$:

$$y\left(\frac{\pi}{6}\right) = 2 \cos\left(\frac{\pi}{2}\right) + \frac{c}{3} \sin\left(\frac{\pi}{2}\right) = 2(0) + \frac{c}{3}(1) = \frac{c}{3}.$$

Set this equal to the given value 1:

$$\frac{c}{3} = 1 \quad \implies \quad c = 3.$$

Step 6 — Write the explicit solution. Back-substitute $c = 3$, so $\frac{c}{3} = 1$:

$$y(t) = 2 \cos(3t) + \sin(3t).$$

Check: $y(0) = 2 \cos 0 + \sin 0 = 2$, and $y(\frac{\pi}{6}) = 2(0) + 1 = 1$. Both boundary conditions are satisfied.

Closing Takeaway

Two questions decide the route every time. *Where do the conditions live?* If they are stacked at one point $t = a$, **reset the clock with $\tau = t - a$** so they land at the origin. If they are *split* between $t = 0$ and a far point, **name the missing zero-data c** and let the far condition solve for it. In both cases the Laplace transform itself never changes — only the bookkeeping that delivers usable data to $t = 0$.